

MARKET BASKET ANALYSIS AND CUSTOMER BEHAVIOR ANALYSIS USING PYTHON



Project: Amazon Customer Shopping Behavior Analysis

Video Link - <https://www.loom.com/share/d938e761af424b5ca8c7919f56082a64>

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PROJECT OVERVIEW

This project analyzes Amazon customer shopping behavior using survey-based customer data. The objective is to understand purchasing patterns, customer satisfaction, recommendation effectiveness, and customer segmentation.

Project Objectives:

- Analyze customer demographics and shopping behavior.
- Identify purchasing patterns and search preferences.
- Segment customers based on shopping behavior.
- Evaluate recommendation effectiveness.
- Generate business recommendations.

STEP 1: IMPORT LIBRARIES

Imported the required Python libraries for data cleaning, analysis, visualization, and machine learning.

- Pandas: Used for data manipulation and analysis.
- NumPy: Used for numerical operations.
- Matplotlib and Seaborn: Used for data visualization.
- Scikit-Learn: Used for customer segmentation through K-Means clustering.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
df = pd.read_csv("Amazon.csv")
df.head()
```

STEP 2: LOAD DATASET

Loaded the Amazon customer shopping behavior dataset into a Pandas Data Frame for analysis.

```
df.head()
```

	Timestamp	age	Gender	Purchase_Frequency	Purchase_Categories	Personalized_Recommendation_Frequency	Browsing_Frequency	Product_Search_Method	Search_
0	2023/06/08 7:50:55 PM GMT+5:30	65	Prefer not to say	Less than once a month	Clothing and Fashion		Yes	Multiple times a day	Keyword
1	2023/06/09 9:37:44 AM GMT+5:30	20	Male	Once a week	Groceries and Gourmet Food;Beauty and Personal...		No	Rarely	Filter
2	2023/06/11 11:26:54 PM GMT+5:30	42	Male	Once a week	Groceries and Gourmet Food;Beauty and Personal...		Sometimes	Few times a week	Keyword
3	2023/06/08 5:17:10 PM GMT+5:30	65	Others	Once a month	Beauty and Personal Care;Clothing and Fashion;...		No	Few times a month	Filter
4	2023/06/11 10:59:30 PM GMT+5:30	45	Female	Once a week	Beauty and Personal Care;Clothing and Fashion;...		Sometimes	Few times a month	NaN

5 rows × 24 columns

Step 3: BASIC EXPLORATION

Performed initial exploration of the dataset to understand its structure, data types, and overall quality. This step helped identify missing values, duplicate columns, and variables requiring preprocessing before analysis.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 800 entries, 0 to 799
Data columns (total 24 columns):
#   Column                                     Non-Null Count  Dtype
---  -
0   Timestamp                                800 non-null    object
1   age                                       800 non-null    int64
2   Gender                                   800 non-null    object
3   Purchase_Frequency                       800 non-null    object
4   Purchase_Categories                      800 non-null    object
5   Personalized_Recommendation_Frequency   800 non-null    object
6   Browsing_Frequency                       800 non-null    object
7   Product_Search_Method                    643 non-null    object
8   Search_Result_Exploration                 800 non-null    object
9   Customer_Reviews_Importance              800 non-null    int64
10  Add_to_Cart_Browsing                     800 non-null    object
11  Cart_Completion_Frequency                 800 non-null    object
12  Cart_Abandonment_Factors                  800 non-null    object
13  Saveforlater_Frequency                    800 non-null    object
14  Review_Left                              800 non-null    object
15  Review_Reliability                       800 non-null    object
16  Review_Helpfulness                       800 non-null    object
17  Personalized_Recommendation_Frequency   800 non-null    int64
18  Recommendation_Helpfulness                800 non-null    object
19  Rating_Accuracy                          800 non-null    int64
20  Shopping_Satisfaction                     800 non-null    int64
21  Service_Appreciation                     800 non-null    object
22  Improvement_Areas                        800 non-null    object
23  transaction                              800 non-null    int64
```

Task 1 – Data Cleaning

Performed data cleaning to improve data quality and prepare the dataset for analysis.

- Removed duplicate records.
- Standardized categorical values.
- Handled missing values.
- Fixed duplicate column names.
- Converted rating columns to numeric format.

Task 1.1: Remove Duplicate Survey Responses

```
] df.duplicated().sum()
```

```
] np.int64(0)
```

Task 1.2: Standardize Categorical Entries

```
] # Standardize Gender  
df['Gender'] = df['Gender'].str.strip().str.title()
```

```
] # Standardize Purchase Frequency  
df['Purchase_Frequency'] = df['Purchase_Frequency'].str.strip().str.title()
```

```
] #Standardize Recommendation Responses  
df['Review_Left'] = df['Review_Left'].str.strip().str.title()  
  
df['Review_Helpfulness'] = df['Review_Helpfulness'].str.strip().str.title()  
  
df['Recommendation_Helpfulness'] = df['Recommendation_Helpfulness'].str.strip().str.title()
```

Task 1.3: Handle Missing Values and Inconsistent Formats

```
df.isnull().sum()
```

```
Timestamp          0  
age                0  
Gender             0  
Purchase_Frequency 0  
Purchase_Categories 0  
Personalized_Recommendation_Frequency 0  
Browsing_Frequency 0  
Product_Search_Method 157  
Search_Result_Exploration 0  
Customer_Reviews_Importance 0  
Add_to_Cart_Browsing 0  
Cart_Completion_Frequency 0  
Cart_Abandonment_Factors 0  
Saveforlater_Frequency 0  
Review_Left        0  
Review_Reliability 0  
Review_Helpfulness 0  
Personalized_Recommendation_Frequency 0  
Recommendation_Helpfulness 0  
Rating_Accuracy    0  
Shopping_Satisfaction 0  
Service_Appreciation 0  
Improvement_Areas  0  
transaction         0  
dtype: int64
```

```

]: # Removing Extra Spaces from Column Names
df.columns = df.columns.str.strip()
df.columns

]: Index(['Timestamp', 'age', 'Gender', 'Purchase_Frequency',
        'Purchase_Categories', 'Personalized_Recommendation_Frequency',
        'Browsing_Frequency', 'Product_Search_Method',
        'Search_Result_Exploration', 'Customer_Reviews_Importance',
        'Add_to_Cart_Browsing', 'Cart_Completion_Frequency',
        'Cart_Abandonment_Factors', 'Saveforlater_Frequency', 'Review_Left',
        'Review_Reliability', 'Review_Helpfulness',
        'Personalized_Recommendation_Frequency', 'Recommendation_Helpfulness',
        'Rating_Accuracy', 'Shopping_Satisfaction', 'Service_Appreciation',
        'Improvement_Areas', 'transaction'],
        dtype='object')

```

Task 1.5: Convert Rating Columns to Numeric Datatype

```

]: # Convert Numerical Rating Columns
rating_cols = ['Customer_Reviews_Importance', 'Rating_Accuracy', 'Shopping_Satisfaction']

for col in rating_cols:
    df[col] = pd.to_numeric(df[col], errors='coerce')

df[rating_cols].dtypes

]: Customer_Reviews_Importance    int64
   Rating_Accuracy                int64
   Shopping_Satisfaction          int64
   dtvce: object

```

Task 2: DESCRIPTIVE BEHAVIOR ANALYSIS

Performed exploratory data analysis (EDA) to understand customer shopping behavior and purchasing patterns.

2.1 Customer Demographics Analysis

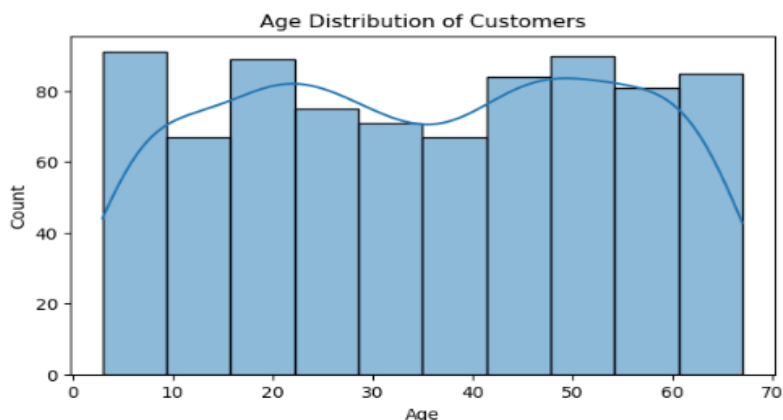
Customer demographics were analyzed using age and gender information.

Age Distribution

```

plt.figure(figsize=(7,4))
sns.histplot(df['age'], bins=10, kde=True)
plt.title('Age Distribution of Customers')
plt.xlabel('Age')
plt.ylabel('Count')
plt.show()

```

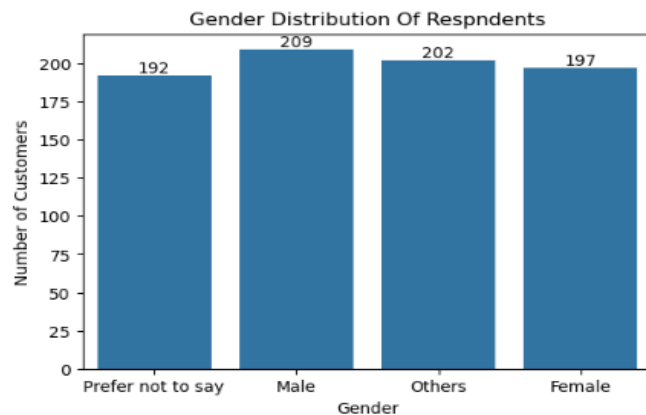


Insight The age distribution shows the spread of Amazon users across different age groups. Most respondents belong to the adult age category, indicating that Amazon's primary customer base consists of active online shoppers.

Gender Distribution

```
plt.figure(figsize=(6,4))
ax = sns.countplot(x='Gender', data=df)
for container in ax.containers:
    ax.bar_label(container)

plt.title('Gender Distribution Of Respdents')
plt.xlabel('Gender')
plt.ylabel('Number of Customers')
plt.show()
```



Insight : This chart shows the proportion of customers across gender categories. Understanding gender distribution helps Amazon design targeted marketing campaigns and personalized recommendations.

2.2 Purchase Frequency Analysis

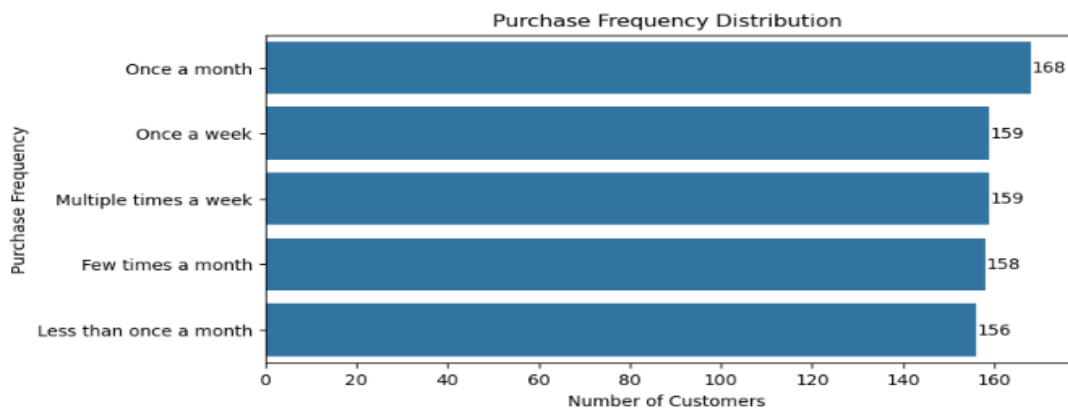
Purchase frequency was analyzed to understand how often customers make purchases on Amazon.

```
plt.figure(figsize=(8,4))

ax = sns.countplot(y='Purchase_Frequency', data=df, order=df['Purchase_Frequency'].value_counts().index)

for container in ax.containers:
    ax.bar_label(container)

plt.title('Purchase Frequency Distribution')
plt.xlabel('Number of Customers')
plt.ylabel('Purchase Frequency')
plt.show()
```



Insight: Purchase frequency helps identify how often customers engage with Amazon. Customers purchasing multiple times per week indicate high engagement and loyalty, while customers purchasing less than once a month may require retention strategies. [1](#)

2.3 Product Search Methods

Customer search behavior was analyzed to understand how shoppers discover products on Amazon.

```
# Identify top browsing methods and most common cart abandonment factors.  
# So we use Product_Search_Method and Cart_Abandonment_Factors.  
plt.figure(figsize=(7,4))  
  
ax = sns.countplot(  
    y='Product_Search_Method',  
    data=df,  
    order=df['Product_Search_Method'].value_counts().index  
)  
for container in ax.containers:  
    ax.bar_label(container)  
  
plt.title('Product Search Methods')  
plt.xlabel('Number of Customers')  
plt.ylabel('Search Method')  
plt.show()
```



Insight: Customer search behavior was analyzed to understand how shoppers discover products on Amazon.

2.4 Cart Abandonment Analysis

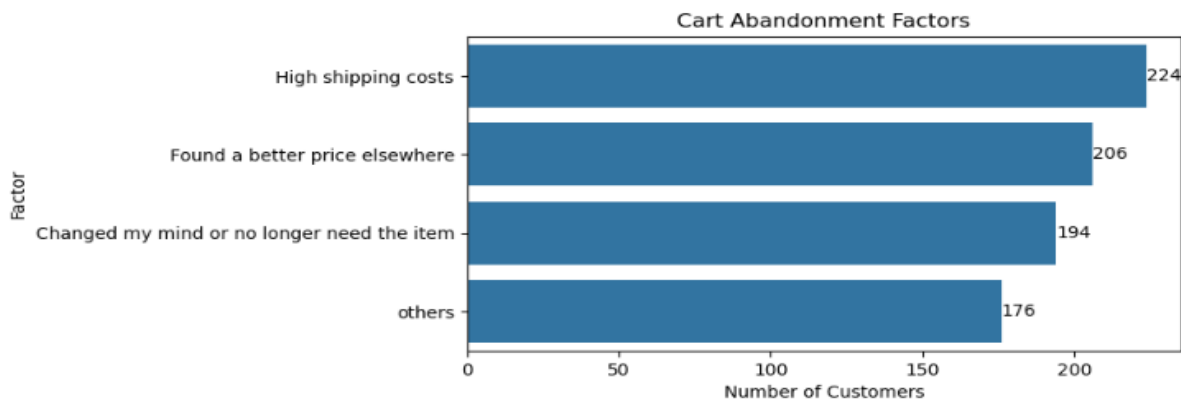
Cart abandonment factors were analyzed to identify the major reasons customers leave products in their carts without completing the purchase. Understanding these factors helps improve conversion rates and customer experience.

```
plt.figure(figsize=(7,3.5))

ax = sns.countplot(
    y='Cart_Abandonment_Factors',
    data=df,
    order=df['Cart_Abandonment_Factors'].value_counts().index)

for container in ax.containers:
    ax.bar_label(container)

plt.title('Cart Abandonment Factors')
plt.xlabel('Number of Customers')
plt.ylabel('Factor')
plt.show()
```



2.5 Summary Statistics

Summary statistics were calculated to understand overall customer satisfaction, recommendation helpfulness, and rating accuracy.

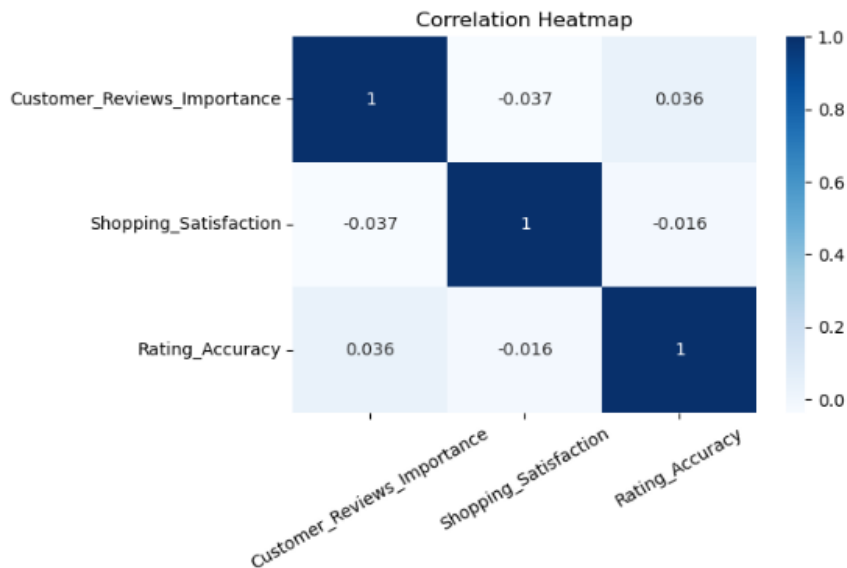
Summary Statistics

```
df[['Customer_Reviews_Importance', 'Shopping_Satisfaction', 'Rating_Accuracy']].describe()
```

	Customer_Reviews_Importance	Shopping_Satisfaction	Rating_Accuracy
count	800.000000	800.000000	800.000000
mean	3.058750	3.01250	2.968750
std	1.408111	1.40706	1.403651
min	1.000000	1.00000	1.000000
25%	2.000000	2.00000	2.000000
50%	3.000000	3.00000	3.000000
75%	4.000000	4.00000	4.000000
max	5.000000	5.00000	5.000000

Insights : The average values for satisfaction, recommendation helpfulness, and rating accuracy are close to the midpoint of the scale, indicating a generally neutral customer perception.

```
plt.figure(figsize=(6,4))
sns.heatmap(df[['Customer_Reviews_Importance','Shopping_Satisfaction','Rating_Accuracy']].corr(),
            annot=True,cmap='Blues')
plt.title('Correlation Heatmap')
plt.xticks(rotation=30)
plt.show()
```



Insight: The variables show very weak correlations with each other, indicating that customer satisfaction, rating accuracy, and review importance behave relatively independently in this dataset.

Task 3: CUSTOMER SEGMENTATION AND PROFILING

Customers were segmented using purchase frequency, shopping satisfaction, and behavioral characteristics.

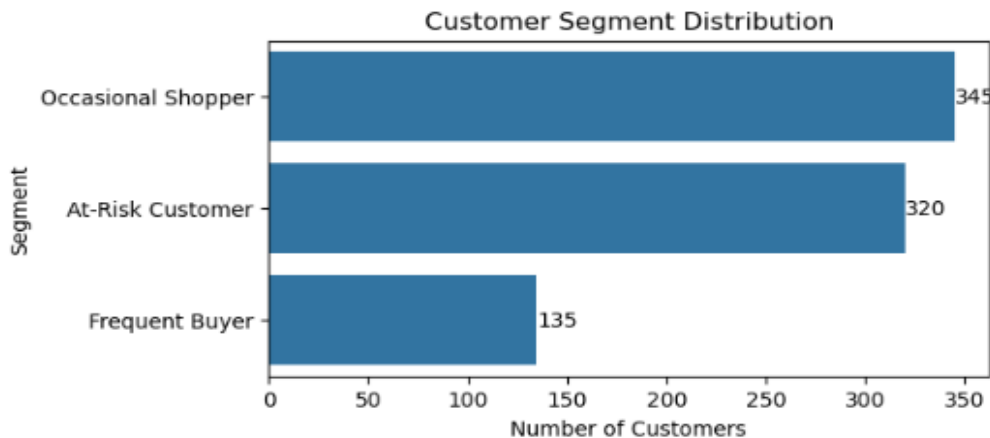
3.1 Rule-Based Customer Segmentation

Customers were classified into Frequent Buyers, Occasional Shoppers, and At-Risk Customers based on purchase frequency and shopping satisfaction levels.

```
# Creating Segment Function
def customer_segment(row):
    # Frequent Buyers
    if (
        row['Purchase_Frequency'] in ['Once A Week', 'Multiple Times A Week']
        and row['Shopping_Satisfaction'] >= 4
    ):
        return 'Frequent Buyer'
    # At-Risk Customers
    elif (
        row['Shopping_Satisfaction'] <= 2
    ):
        return 'At-Risk Customer'
    # Occasional Shoppers
    else:
        return 'Occasional Shopper'
df['Customer_Segment'] = df.apply(customer_segment, axis=1)
```

```
plt.figure(figsize=(6,3))
ax = sns.countplot(
    y='Customer_Segment',
    data=df,
    order=df['Customer_Segment'].value_counts().index)

for container in ax.containers:
    ax.bar_label(container)
plt.title('Customer Segment Distribution')
plt.xlabel('Number of Customers')
plt.ylabel('Segment')
plt.show()
```



Insight: Most customers belong to the Occasional Shopper segment, while Frequent Buyers represent highly engaged customers and At-Risk Customers require retention-focused strategies.

3.2 Demographic and Behavioral Analysis

The customer segments were analyzed based on shopping satisfaction, rating accuracy, and review importance to identify behavioral differences.

```
# Age by Segment
df.groupby('Customer_Segment')['age'].mean()
```

```
Customer_Segment
At-Risk Customer    35.790625
Frequent Buyer     34.607407
Occasional Shopper  34.814493
Name: age, dtype: float64
```

```
# Gender by Segment
pd.crosstab(df['Customer_Segment'], df['Gender'])
```

	Gender	Female	Male	Others	Prefer Not To Say
Customer_Segment					
At-Risk Customer		72	93	77	78
Frequent Buyer		30	35	31	39
Occasional Shopper		95	81	94	75

Insights: Frequent Buyers show higher satisfaction levels, while At-Risk Customers report lower satisfaction scores.

3.3 K-Means Clustering

K-Means clustering was applied to identify customer groups based on shopping behavior and review-related factors.

```
#Selecting Features
#Using behavioral variables
features = [
    'Purchase_Frequency_Num',
    'Shopping_Satisfaction',
    'Rating_Accuracy'
]

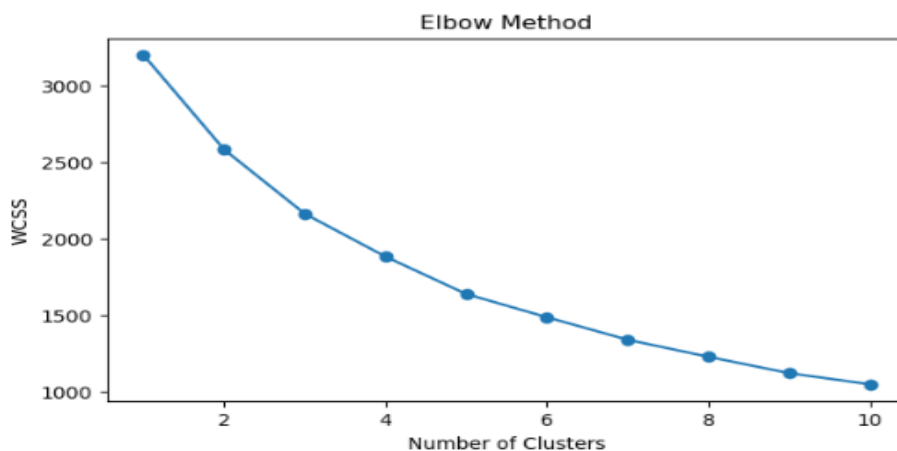
# Scaling Data
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_scaled = scaler.fit_transform(df[features])
```

3.4 Elbow Method

The Elbow Method was applied to determine the optimal number of customer clusters.

```
from sklearn.cluster import KMeans
wcss = []
for i in range(1, 11):
    kmeans = KMeans(
        n_clusters=i,
        random_state=42,
        n_init=10)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

plt.figure(figsize=(7,4))
plt.plot(range(1,11), wcss, marker='o')
plt.title('Elbow Method')
plt.xlabel('Number of Clusters')
plt.ylabel('WCSS')
plt.show()
```

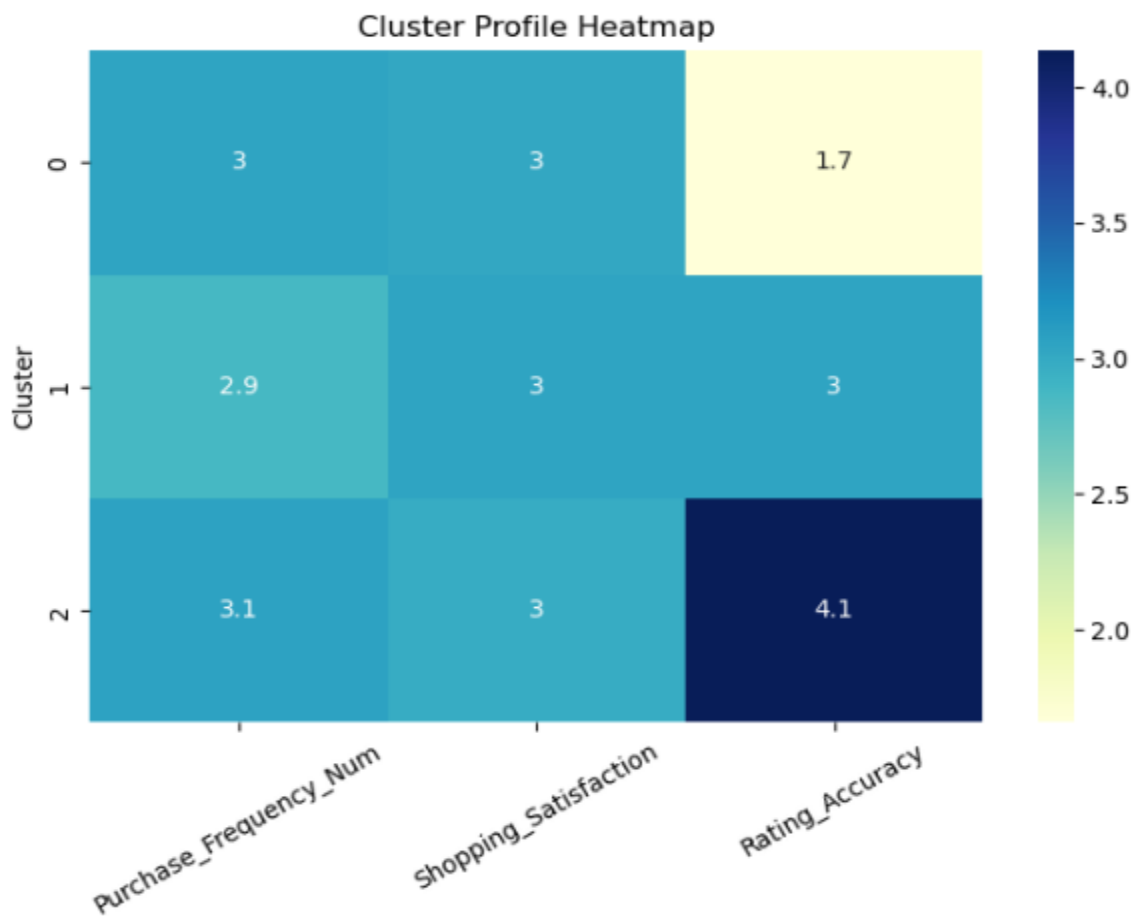


Insights : The Elbow Method suggested three clusters as the optimal grouping, balancing model simplicity and cluster separation.

3.5 Cluster Profiling

After clustering, the average values of key behavioral variables were analyzed to understand the characteristics of each cluster. This profiling helps identify distinct customer groups and their preferences.

```
cluster_profile
plt.figure(figsize=(8,5))
sns.heatmap(
    cluster_profile,
    annot=True,
    cmap='YlGnBu'
)
plt.title('Cluster Profile Heatmap')
plt.xticks(rotation=30)
plt.show()
```



Insights: The cluster analysis reveals distinct customer groups that differ primarily in review importance, rating accuracy, and recommendation helpfulness, highlighting diverse customer preferences and shopping behaviors.

Task 4: Recommendation and Review Insights

Customer reviews and recommendation-related factors were analyzed to understand their impact on shopping satisfaction, rating accuracy, and customer engagement.

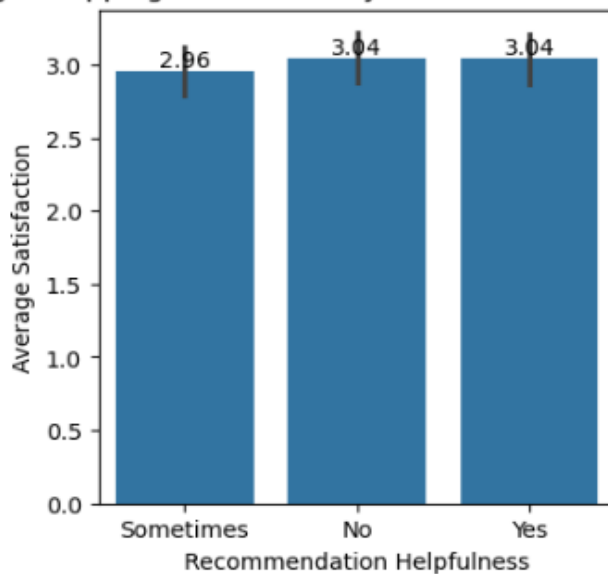
4.1 Recommendation Helpfulness and Shopping Satisfaction

The relationship between recommendation helpfulness and customer satisfaction was analyzed.

```
plt.figure(figsize=(4,4))
ax = sns.barplot(x='Recommendation_Helpfulness',y='Shopping_Satisfaction', data=df, estimator='mean')
for container in ax.containers:
    ax.bar_label(container, fmt='%.2f')

plt.title('Average Shopping Satisfaction by Recommendation Helpfulness')
plt.xlabel('Recommendation Helpfulness')
plt.ylabel('Average Satisfaction')
plt.show()
```

Average Shopping Satisfaction by Recommendation Helpfulness

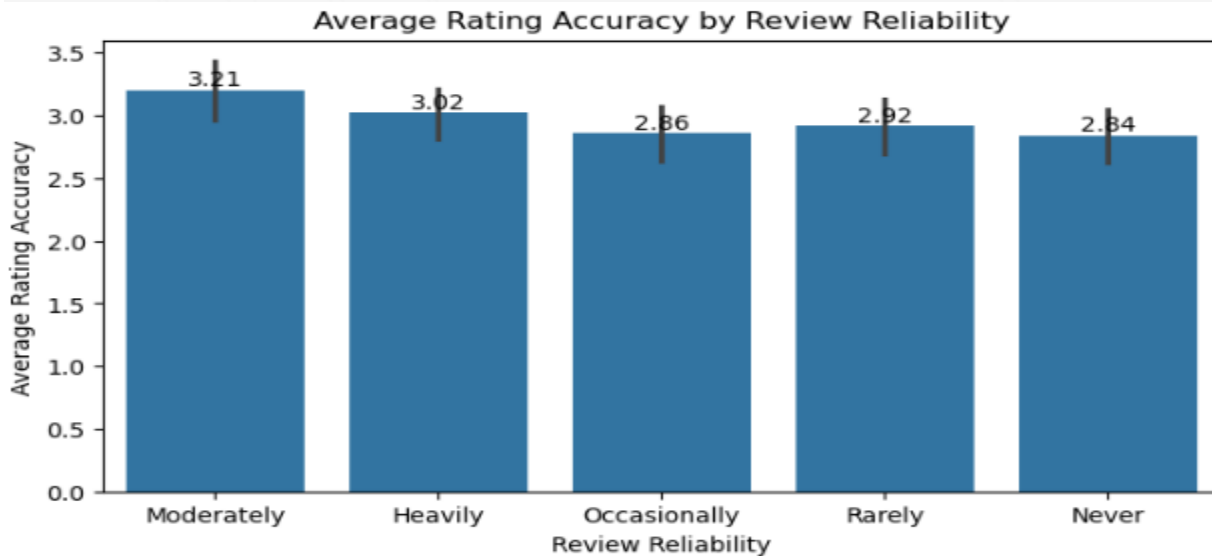


Insights: Customers who find recommendations helpful tend to report slightly higher shopping satisfaction.

4.2 Review Reliability and Rating Accuracy

Review reliability was analyzed to understand its impact on perceived rating accuracy.

```
plt.figure(figsize=(8,4))
ax = sns.barplot(x='Review_Reliability',y='Rating_Accuracy',data=df,estimator='mean')
for container in ax.containers:
    ax.bar_label(container, fmt='%.2f')
plt.title('Average Rating Accuracy by Review Reliability')
plt.xlabel('Review Reliability')
plt.ylabel('Average Rating Accuracy')
plt.show()
```



Insights: Higher review reliability is associated with better rating accuracy and increased customer trust.

4.3 Review Helpfulness and Shopping Satisfaction

The impact of review helpfulness on customer satisfaction was examined.

```
plt.figure(figsize=(6,4))
ax = sns.barplot(x='Review_Helpfulness',y='Shopping_Satisfaction',data=df,estimator='mean')
for container in ax.containers:
    ax.bar_label(container, fmt='%.2f')
plt.title('Shopping Satisfaction by Review Helpfulness')
plt.xlabel('Review Helpfulness')
plt.ylabel('Average Satisfaction')
plt.show()
```

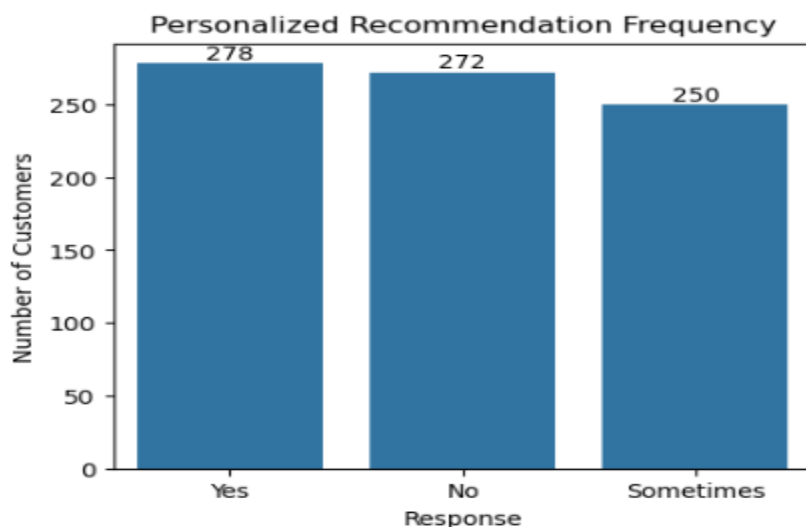


Insights: Helpful reviews positively influence customer satisfaction and purchase confidence.

4.4 Personalized Recommendation Engagement

Customer engagement with personalized recommendations was analyzed to understand recommendation usage patterns.

```
plt.figure(figsize=(5,4))
ax = sns.countplot(
    x='Personalized_Recommendation_Frequency',
    data=df,
    order=df['Personalized_Recommendation_Frequency'].value_counts().index)
for container in ax.containers:
    ax.bar_label(container)
plt.title('Personalized Recommendation Frequency')
plt.xlabel('Response')
plt.ylabel('Number of Customers')
plt.show()
```



Insights: Personalized recommendations are widely used, indicating strong customer engagement with Amazon's recommendation system.

4.5 Actionable Insights for Improving Amazon's Recommendation System

1. *Customers who find recommendations helpful report higher shopping satisfaction. Improving recommendation relevance can enhance customer experience.*
2. *Reliable reviews increase trust in product ratings. Products with trusted reviews should be prioritized in recommendations.*

- 3. Helpful customer reviews positively influence shopping satisfaction. Amazon should highlight useful reviews for customers.*
- 4. Personalized recommendations improve customer engagement. Expanding personalization can increase customer interaction.*
- 5. Customer purchase history and browsing behavior should be used to improve recommendation accuracy.*
- 6. Products with highly rated and reliable reviews should receive greater visibility in recommendation sections.*

TASK 5: VISUALIZATION AND REPORTING

Visualizations were created throughout the project to analyze customer behavior, shopping patterns, customer segments, and recommendation effectiveness. The charts helped identify key trends and supported data-driven business recommendations.

5.1 Dashboard Summary

The analysis revealed important insights regarding customer shopping behavior, satisfaction levels, recommendation effectiveness, and customer segmentation.

Key Findings :

- 1. Most customers purchase products once a month or once a week, indicating regular shopping activity.*
- 2. Keyword search is the most preferred method for discovering products on Amazon.*
- 3. High shipping costs are the leading cause of cart abandonment.*
- 4. Frequent Buyers exhibit higher satisfaction levels than At-Risk Customers.*
- 5. K-Means clustering identified distinct customer groups with different shopping behaviors.*
- 6. Reliable and helpful reviews positively influence customer trust and satisfaction.*
- 7. Personalized recommendations contribute to customer engagement and shopping experience.*

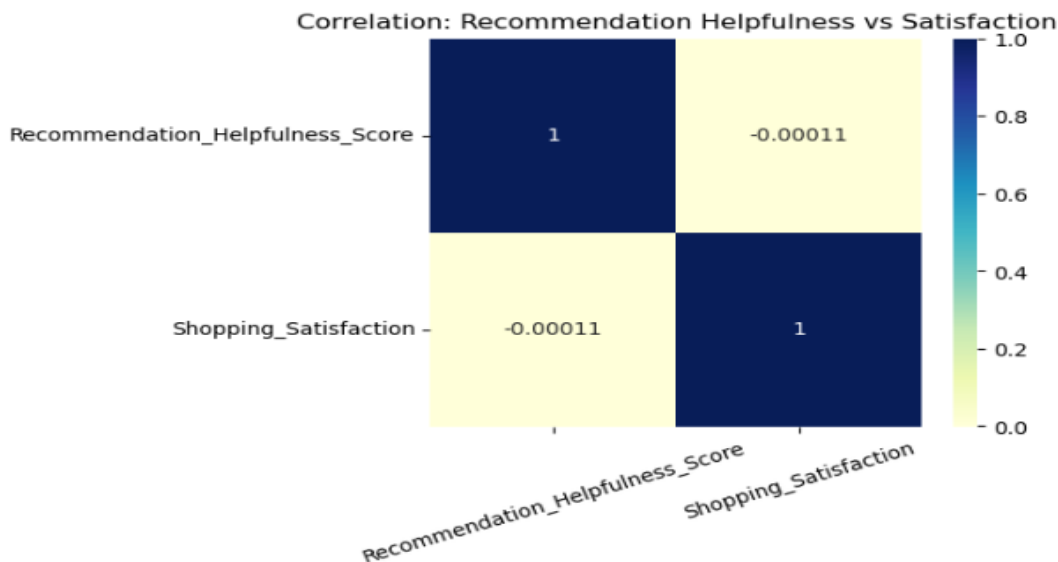
Business Recommendations

1. Improve recommendation accuracy using customer browsing and purchase history.
2. Reduce cart abandonment through shipping cost optimization and targeted offers.
3. Promote products with highly trusted reviews.
4. Expand personalized recommendation strategies to increase customer engagement.
5. Develop retention campaigns for At-Risk Customers.

Conclusion

This project successfully analyzed Amazon customer shopping behavior using data cleaning, exploratory data analysis, customer segmentation, clustering, and recommendation analysis. The findings revealed important purchasing patterns, customer segments, and factors influencing satisfaction and engagement. These insights can help improve recommendation systems, reduce cart abandonment, and enhance the overall customer experience.

```
correlation_data = df[['Recommendation_Helpfulness_Score', 'Shopping_Satisfaction']].corr()
plt.figure(figsize=(5,4))
sns.heatmap(correlation_data, annot=True, cmap='YlGnBu')
plt.title('Correlation: Recommendation Helpfulness vs Satisfaction')
plt.xticks(rotation=20)
plt.show()
```



Thankyou!